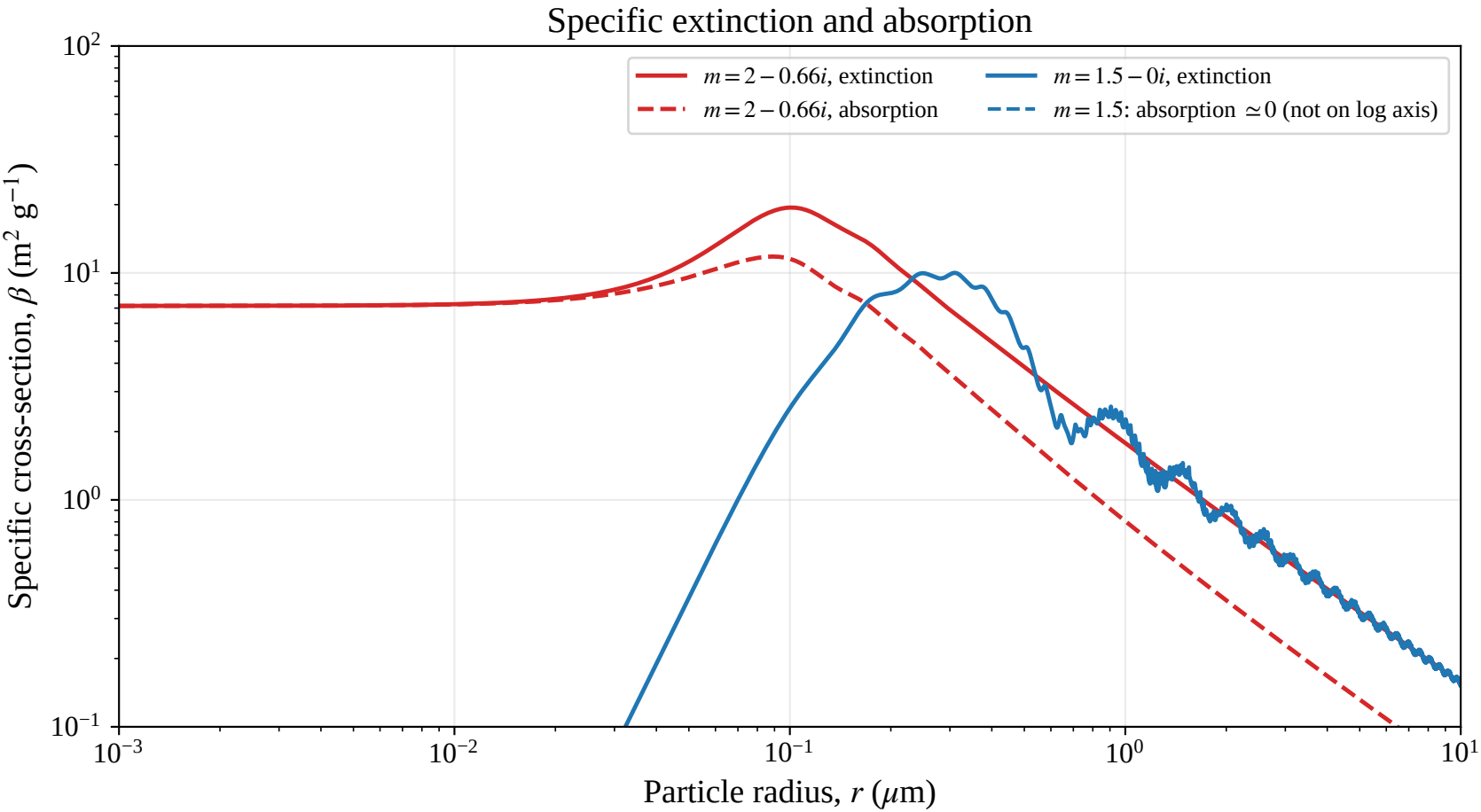$\lambda = 0.55\,\mu$m; $\rho_p = 1\,\mathrm{g\,cm^{-3}}$ (assumed density)